

Locality of $\text{Tr } \phi^3$ tree amplitudes from hidden zeros: a crossing forces one extra chord

Working note — J. Valenzuela

June 9, 2026

Abstract

We prove (modulo one clearly-isolated input, the exhaustiveness of the K -zero survivor catalog of notes 25/30) that any chord diagram surviving all hidden-zero conditions has its arrangement *fully triangulated*, and that a fully triangulated arrangement with p crossings has exactly $m = n - 3 + p$ chords. Since a $\text{Tr } \phi^3$ term carries exactly $n - 3$ chords, a surviving term has $p = 0$: it is a triangulation. Thus locality emerges. The counting step is rigorous; the geometric step is Jony's descent, stated here as a terminating induction.

1 Statement and strategy

Work in a convex n -gon, vertices $1, \dots, n$ cyclic. A *diagram* M is a set of chords (non-adjacent vertex pairs). Two chords *cross* iff their endpoints interleave. A *triangulation* is a maximal non-crossing set; it has $n - 3$ chords. A $\text{Tr } \phi^3$ term is a diagram of exactly $n - 3$ chords.

Theorem 1. *Let M survive all K -zero conditions and let p be its number of crossings. Then $|M| = n - 3 + p$. In particular a surviving term ($|M| = n - 3$) has $p = 0$ and is a triangulation.*

The proof has two independent halves:

- (I) **Counting (§2, rigorous).** If every bounded face of the arrangement is a triangle, then $m = n - 3 + p$.
- (II) **Geometry (§5).** Survival forces every bounded face to be a triangle. This is the descent.

2 The counting lemma

Draw the chords of M as straight segments; perturb to general position so no three are concurrent (the survival question is combinatorial, so this is no loss). Let p be the number of crossing pairs; each is a distinct interior crossing point.

Lemma 1. *If every bounded face of the arrangement is a triangle, then $m = n - 3 + p$.*

Proof. The planar subdivision has $V = n + p$ vertices (corners plus crossing points). Its edges are the n boundary edges together with the chord segments: a chord crossed c times is cut into $c + 1$ pieces, and $\sum_{\text{chords}} c = 2p$, so there are $m + 2p$ chord segments; hence $E = n + m + 2p$. Euler's formula $V - E + F = 2$ gives $F = m + p + 2$, so the number of bounded faces is

$$F_{\text{int}} = F - 1 = m + p + 1.$$

Summing face degrees, each bounded face being a triangle and the outer face having the n boundary edges,

$$3(m + p + 1) + n = \sum_f \deg f = 2E = 2(n + m + 2p),$$

which simplifies to $m = n - 3 + p$. □

Remark 1. Checks: a triangulation has $p = 0$, $m = n - 3$ and $n - 2$ triangular faces. The two diagonals of a square have $p = 1$ and $m = 2 = n - 3 + p$, with four triangular faces. Both verified by direct face construction.

So everything reduces to (II): *survival triangulates every face*. Then $m = n - 3 + p$, and $m = n - 3 \Rightarrow p = 0$.

3 The survival conditions

For a contiguous arc (“green region”) $W = (i, \dots, i + k)$ with closing chord $(i, i + k)$, the K -zero survivor catalog (notes 25, 30) gives, for a surviving diagram, the dichotomy at W :

- (P1) some interior vertex of W carries a chord *out* of W (a chord “shoots out”); *or*
- (P2) the closing chord $(i, i + k)$ extends to a $(k+2)$ -gon (a triangle/cap is built *on top* of it on the far side).

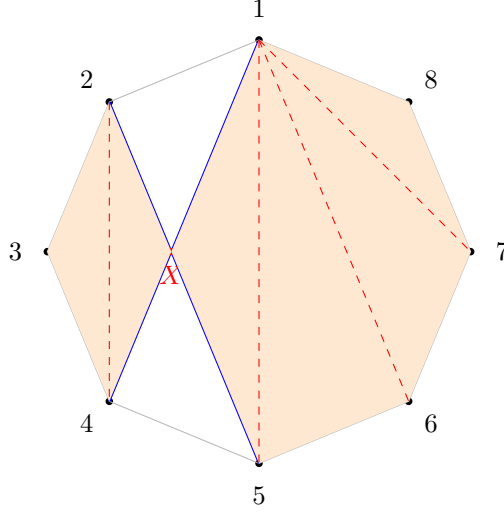
The $k = 1$ instance is: *every boundary edge lies in a triangle*. We take as our single **Input** the established fact that this dichotomy is *exhaustive*: a diagram with no chord “outside the catalogued cases” at W and no extension at W is killed. This is the content of notes 25/30 (“anything with a chord outside these cases survives”); we use it as a black box.

Lemma 2 (Face Lemma). *If M survives all K -zeros, every bounded face of its arrangement is a triangle.*

For a face bounded entirely by full chords/edges (no crossing point on its boundary), the face is an empty sub-polygon; if it had ≥ 4 sides it would be a window with sealed interior and no extension, hence killed. So such faces are triangles. The faces incident to a crossing are handled by the descent below.

4 The simplest crossing

Consider the minimal crossing pair $X_{i-1,j-1}$ and X_{ij} (offset by one), e.g. $(1, 4)$ and $(2, 5)$ in the octagon. Their four endpoints 1, 2, 4, 5 include two adjacent pairs $(1, 2)$ and $(4, 5)$, so only two arcs are non-trivial.



The two blue chords cross at X , creating four bounded faces: two triangles $(1, 2, X)$ and $(4, 5, X)$ and two non-triangular faces, the quadrilateral $(2, 3, 4, X)$ and the hexagon $(5, 6, 7, 8, 1, X)$ (shaded). By the Face Lemma each shaded face must be triangulated. The quadrilateral forces one chord (e.g. $(2, 4)$); the hexagon forces three (e.g. $(1, 5)$, $(1, 6)$, $(1, 7)$). Counting: 2 original + 1 + 3 = 6 = $n - 2$ chords, with $p = 1$, exactly $m = n - 3 + p$. A term has $n - 3 = 5 < 6$ chords, so it *cannot* carry this crossing and survive.

Note the role of the $k = 1$ condition even on the adjacent arcs: edge $(2, 3)$ starts in no triangle, so it alone already forces a chord into the quadrilateral. “Even P1 builds triangles”: any chord introduced to satisfy a window becomes a side of new faces that must themselves be triangulated — one never escapes the triangulation obligation, one only relocates it.

5 The descent (general case)

Let M survive and contain a crossing $(i, j), (r, s)$, $i < r < j < s$. Define on each active arc the measure

$$\Phi(\text{arc}) = \#\{\text{polygon vertices strictly interior to the arc}\}.$$

Anchor a K -zero whose green region is the arc. By the Input dichotomy, survival gives:

- **P2**: the arc is capped by a triangle on its closing chord; recurse on the strictly shorter sub-arc. Φ drops by at least 1.
- **P1**: a chord shoots out from an interior vertex v of the arc. Choosing the green region across a current gap (no chord spans it) makes this chord *new*; it splits the arc at v into $[a, v]$ and $[v, b]$, each of strictly smaller Φ .

Φ is a non-negative integer that strictly decreases at every step; the base case $\Phi \leq 1$ is a triangle. Hence the descent terminates, and on termination every bounded face is a triangle. This proves the Face Lemma 2; with Lemma 1 it proves Theorem 1.

Remark 2 (the new crossings do not restart the descent). A P1 chord may cross existing chords — raising p and, via $m = n - 3 + p$, the final chord count — but it still strictly lowers Φ on the active arc. So crossings accumulate into the “+ p ” while the descent marches monotonically to triangles. This is the precise form of “the distance between vertices shrinks every step.”

6 Equivalent formulation: every survivor contains a triangulation

The descent builds, among the chords of M , a maximal non-crossing family T (the P2 caps and the non-crossing P1 chords) that triangulates the polygon: $|T| = n - 3$ and $T \subseteq M$.

Corollary 1. *A surviving diagram M contains a triangulation T . If M also has a crossing, then some crossing chord lies in $M \setminus T$ (a triangulation is crossing-free), so $|M| \geq |T| + 1 = n - 2$.*

This is the clean structural statement: *survival* $\Rightarrow$ *a triangulation sits inside the diagram*, and a crossing is necessarily one chord *beyond* it. It is equivalent to Theorem 1 via Lemma 1.

7 Status: what is proved and what is assumed

- **Rigorous:** the counting lemma 1 ($m = n - 3 + p$) and the reduction “fully triangulated + $m = n - 3 \Rightarrow$ triangulation”; the worked simplest case; the termination of the descent via the integer measure Φ .
- **Assumed (Input):** the K -zero survivor catalog of notes 25/30 is *exhaustive* — at each green region a survivor satisfies P1 or P2 and nothing else, and when neither holds the coefficient vanishes. This is the established hidden-zero analysis; we use it as a black box. The Face Lemma rests on it.
- **To write carefully (bookkeeping, not new ideas):** that the P1 chord is always genuinely new (choose the green region across a current gap) and interior to the active arc, so Φ provably drops; and that the non-crossing chords built tile the polygon into a single triangulation T (Corollary 1) without boundary double-counting.

Modulo the Input, the argument is complete: survival triangulates every face, Euler turns that into $m = n - 3 + p$, and the chord budget $n - 3$ forces $p = 0$. Locality emerges.